

Minor Project



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NURIA

A NEW ERA OF ROBOTIC NURSE WITH INTELLIGENT PATIENT MONITORING SERVICE

Name: **Ms. Akshara Venkatraman, Ms. Devika U P**

Department of CINTEL, SRM Institute of Science and Technology, KTR campus

Registration No. : RA2311026011079 , RA2311026011089

Project Supervisor: **Dr. Prithvi Krishna Chittoor**, Assistant Professor, Department of CINTEL, SRM

Institute of Science and Technology, KTR campus

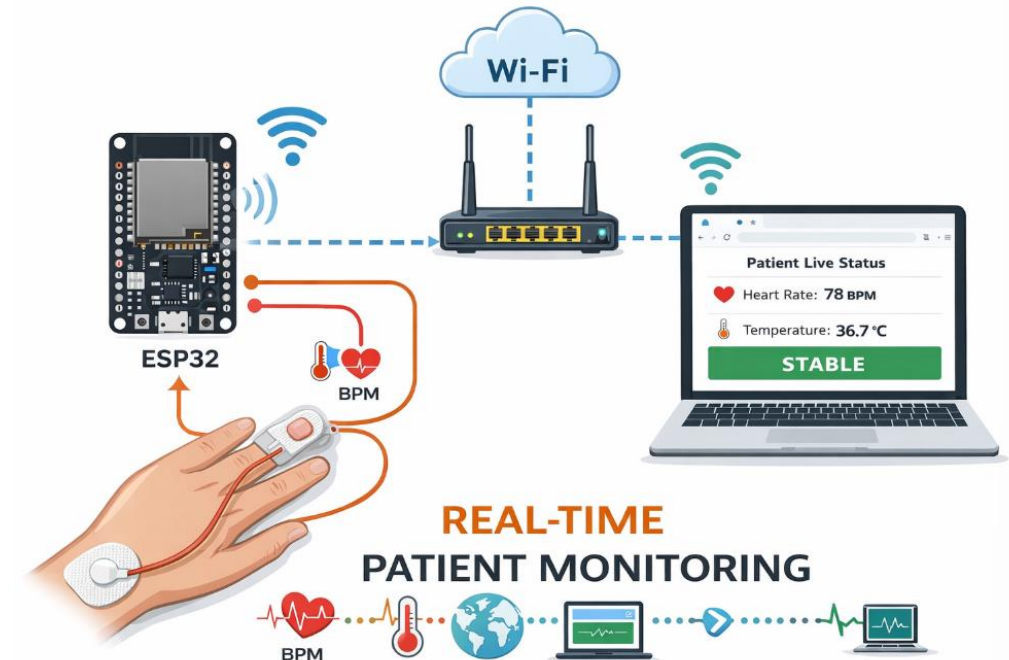
Overview of Presentation

- **Abstract**
- **Introduction**
- **Literature Survey**
- **Research Objective**
- **Project SDG**
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Abstract



This project implements a real-time patient monitoring system using an ESP32-based IoT platform. The system collects vital parameters such as heart rate and body temperature and displays them on a live web dashboard over WiFi. The patient's condition is automatically classified as Stable or Emergency based on predefined thresholds. The solution demonstrates a low-cost, portable, and scalable approach for real-time health monitoring and quick emergency identification.



Introduction to NURIA

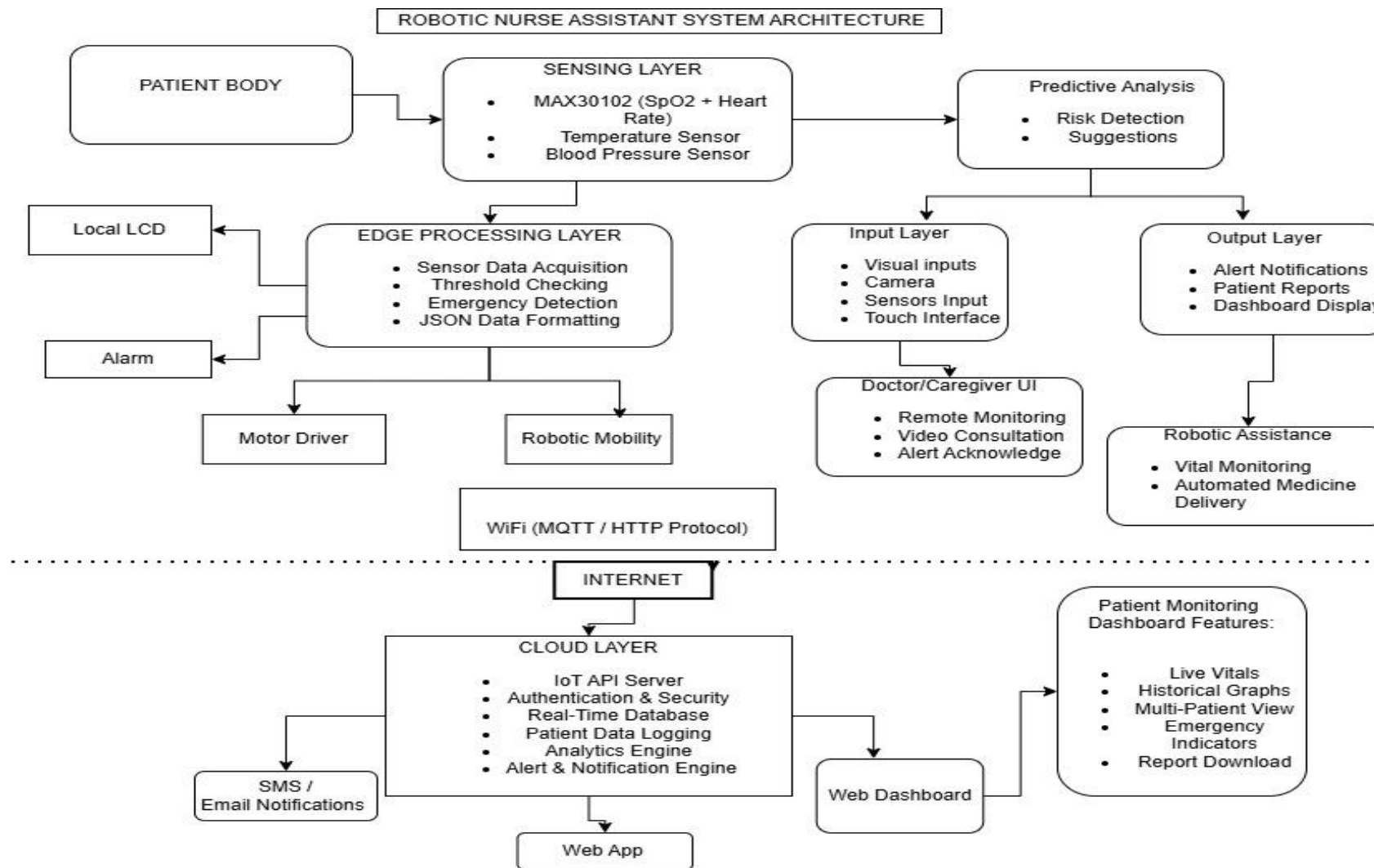


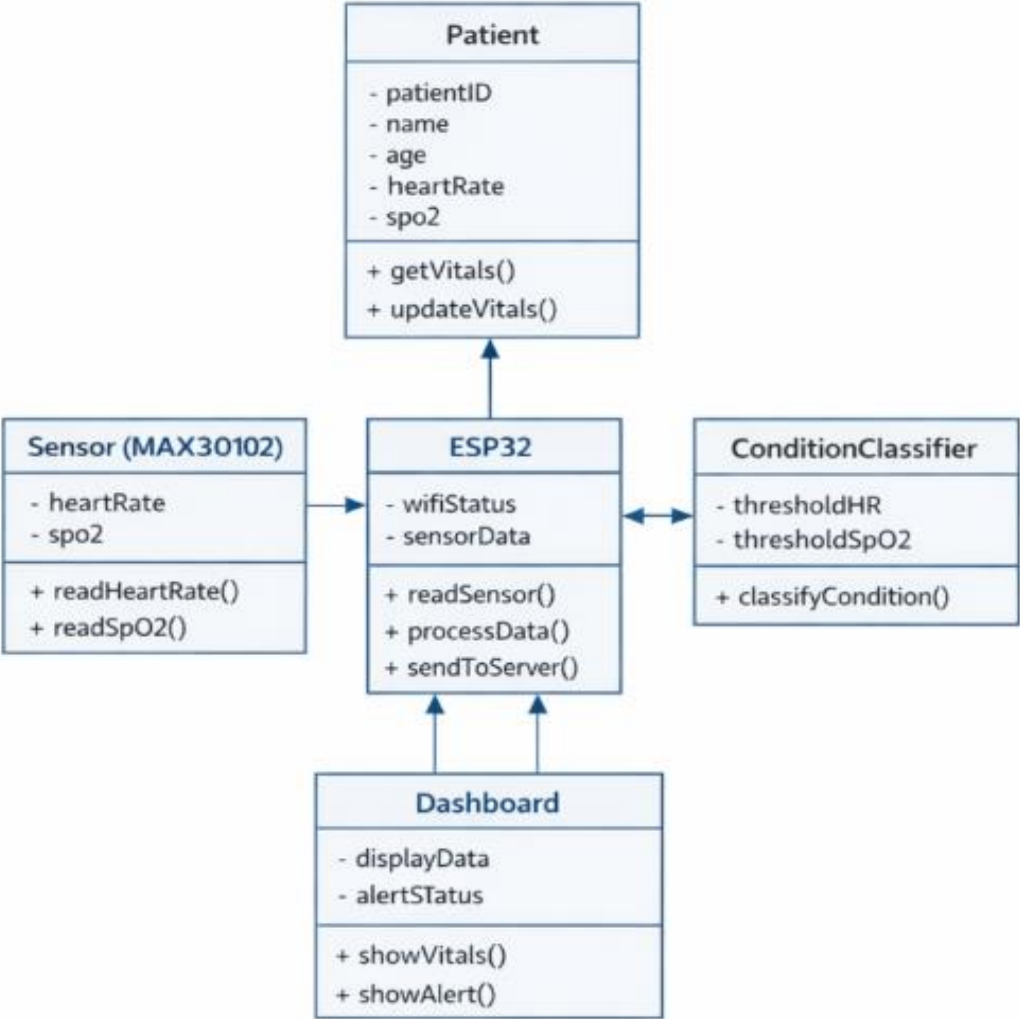
- Hospitals worldwide sometimes face nurse shortages, heavy workload, fatigue and burnout, leading to delays in patient monitoring and medicine administration.
- NURIA is designed to act as a supportive robotic nurse that continuously monitors patient vitals, assists nurses during busy hours, and ensures timely delivery of medicines, improving patient safety and hospital efficiency.



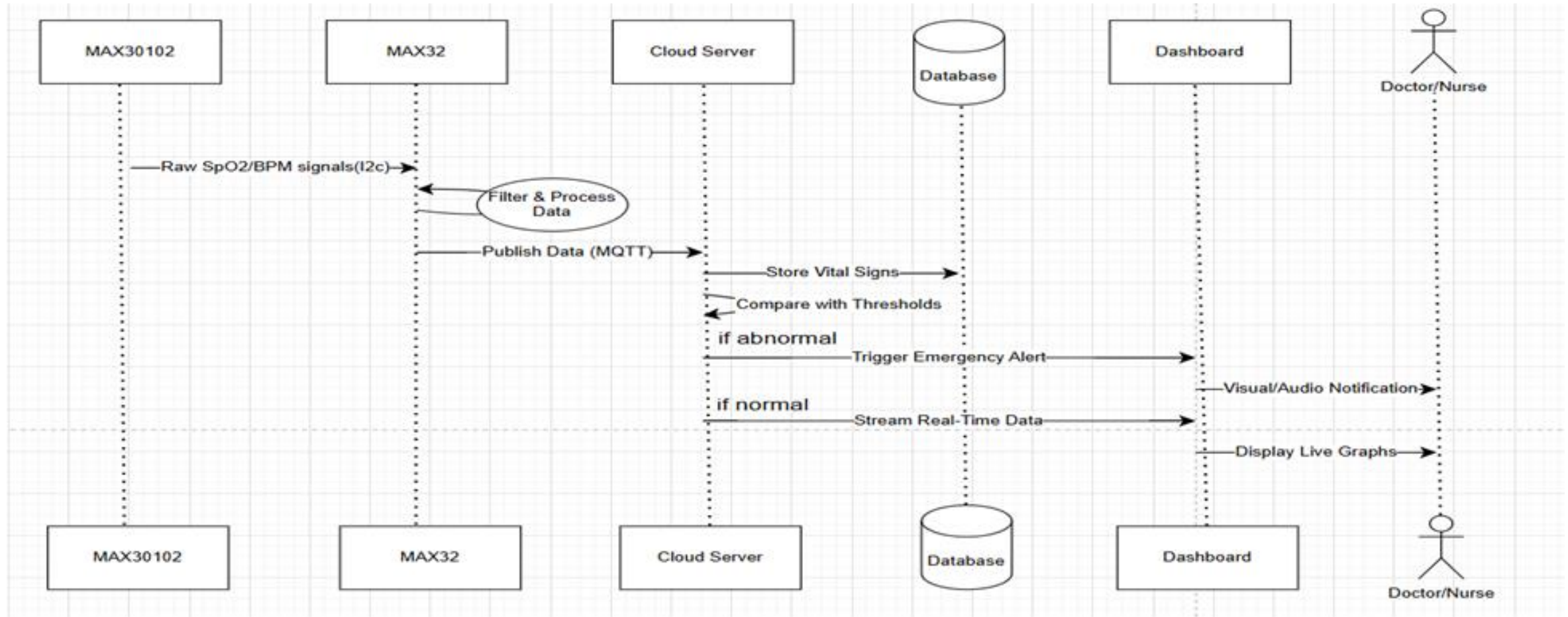
Fig. 1. Robot Assuring Old Man
Source: www.shutterstock.com

System Architecture

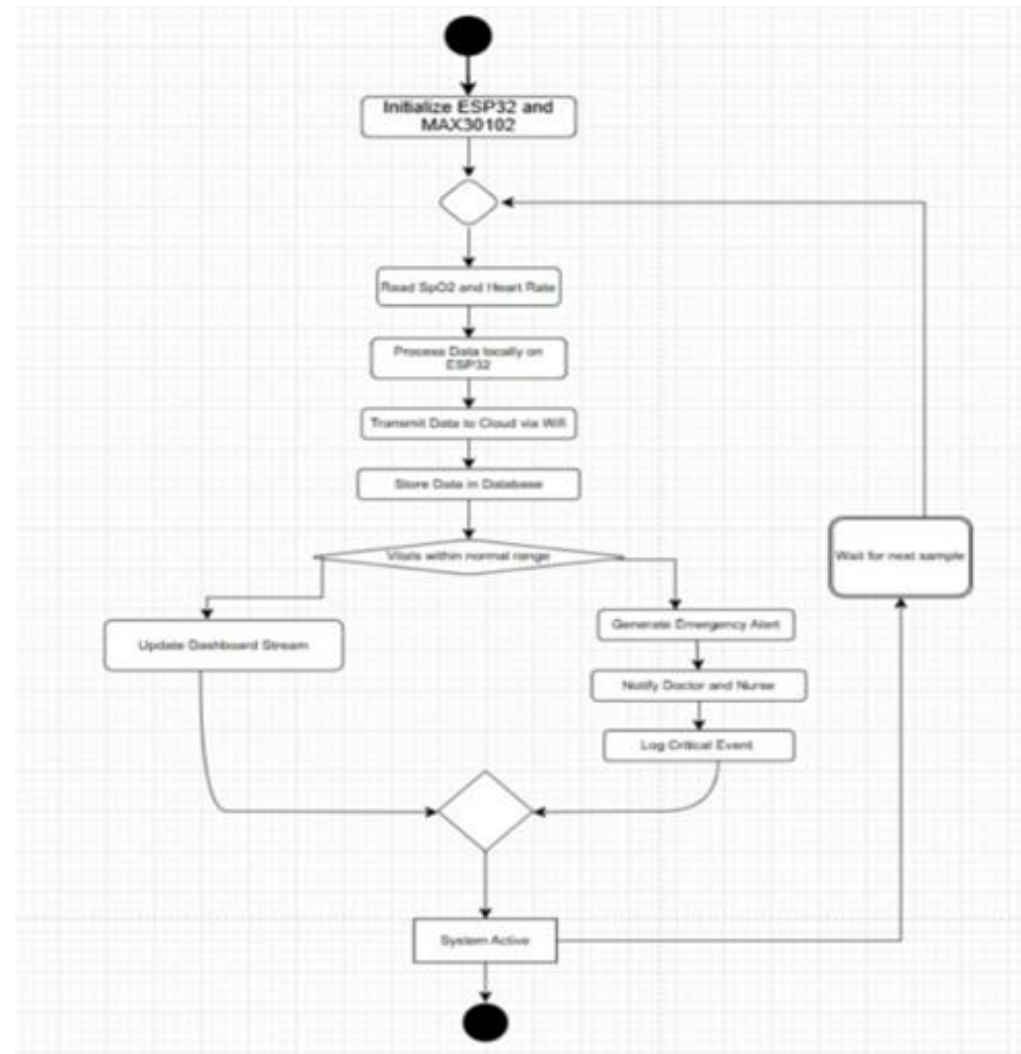




Sequence Diagram



Activity Diagram



https://drive.google.com/file/d/1HXXrX4yZ1OfmXi36uCmzUeNya60QDo5s/view?usp=drive_link



Literature Survey (1/5)



Author(s)	Title	Journal	Inference
Yee Wei Lim et al., 2024	An Assessment of an Inpatient Robotic Nurse Assistant: A Mixed-Method Study	Journal of Medical Systems	- Mixed-method clinical evaluation of a robotic nurse assistant that supports vital signs measurement and medication. Real human subject data, measures vitals + med delivery + HRI (human–robot interaction). Limited autonomy & interactivity; early prototype with scalability issues.
C. Lundberg et al., 2022	Robotic Nursing Assistant Applications and Human Subject Tests with a Mobile Robotic Platform	MDPI Robotics	- Tests a mobile robotic platform performing basic nursing tasks like object fetching and temperature measurement . Real human subjects, practical feasibility of task execution. No full clinical deployment, limited to basic tasks.
K. Wada & T. Shibata, 2008	Robot therapy for elderly care	IEEE Engineering in Medicine and Biology Magazine	- Evaluates robots used to provide social or therapeutic assistance in elderly care. Shows benefits of robotic assistance for patient well-being. No automation of medical tasks or monitoring.

Literature Survey (2/5)



Author(s)	Title	Journal	Inference
Justinas Miseikis et al., 2020	Lio-A personal robot assistant for human-robot interaction and care applications	IEEE Robotics and Automation Letters	• Mobile robot with manipulator arm integrating vision, audio, laser, and ultrasound sensors • Demonstrates autonomous navigation and task scheduling in real deployments. • No dedicated vital signs monitoring.
Mohammad N. Saadatzi et al., 2020	Acceptability of using a robotic nursing assistant in health care environments: Experimental pilot study	Journal of Medical Internet Research (JMIR)	• Evaluates Adaptive Robotic Nurse Assistant (ARNA) for ambulatory tasks. • Reports high acceptance and practical design insights. • Does not include advanced vitals analytics or clinical decision support.
Paolo Fiorini et al., 2017	Robot-Assisted Healthcare Services	IEEE Reviews in Biomedical Engineering	• Survey of service robots in hospitals for delivery, monitoring, and telepresence. • Highlights improved workflow efficiency. • Limited real-time physiological monitoring integration.
Oussama Khatib et al., 2015	Human-Centered Robotics for Physical Assistance in Care Settings	IEEE International Conference on Robotics	• Focuses on manipulation and mobility assistance. • Demonstrates safe patient support mechanisms. • No embedded health monitoring module.

Literature Survey (3/5)



Author(s)	Title	Journal	Inference
E. Broadbent et al., 2014	Robots with interactive display interfaces for healthcare monitoring and patient communication	International Journal of Social Robotics	- Robots communicate patient info via display interfaces. - Practical HRI design for healthcare settings. - Does not automate vitals or medicine delivery.
C. Chen et al., 2020	Artificial intelligence enabled healthcare robots for patient monitoring and assistance	IEEE Access	- AI algorithms integrated with robot platform for monitoring and data analysis. - Real AI and sensor integration. - Limited physical manipulation and no medicine delivery.
Tatsuya Shibata et al., 2012	Therapeutic Robot for Dementia Care (PARO)	Proceedings of the IEEE	- PARO robotic seal improves emotional engagement in elderly patients. - Clinically validated for dementia therapy - Focuses on therapy, not medical monitoring.
Hongbin Zha et al., 2019	Autonomous Service Robot for Hospital Logistics	IEEE Access	- Mobile robot system for hospital delivery and logistics automation. - Improves operational efficiency. - No patient health monitoring functions.

Literature Survey (4/5)



Author(s)	Title	Journal	Inference
Antonio Bicchi et al., 2018	Safe Physical Human–Robot Interaction in Healthcare Robotics	IEEE Transactions on Robotics	- Discusses compliant actuation and safe manipulation techniques for assistive robots. - Ensures safety in patient interaction. - Does not address medical data processing.
Shuzhi Sam Ge et al., 2019	AI-Based Service Robots for Healthcare Assistance	IEEE Transactions on Automation Science and Engineering	- Integrates AI decision systems with service robots for adaptive task execution. - Autonomous scheduling and navigation achieved. - Limited physiological signal analytics.
Cynthia Breazeal et al., 2016	Social Robots for Health and Elder Care Applications	IEEE Robotics and Automation Magazine	- Explores socially assistive robots in elderly care environments. - Enhances engagement and therapy compliance. - Lacks medical sensor integration.
Daniele Nardi et al., 2014	Assistive Robots for Ambient Assisted Living	IEEE International Conference on Advanced Robotics	- Integration of assistive robots in smart healthcare environments. - Supports elderly independent living. - Limited real-time health data analytics.

Literature Survey (5/5)

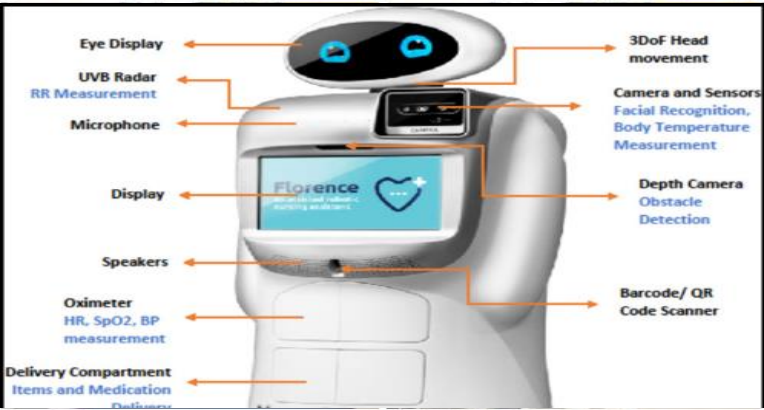


Fig. 2. Image of Robotic Nurse Assistant, Florence, and its Functions



Fig. 3. robotic arm using target frame for fetching task.

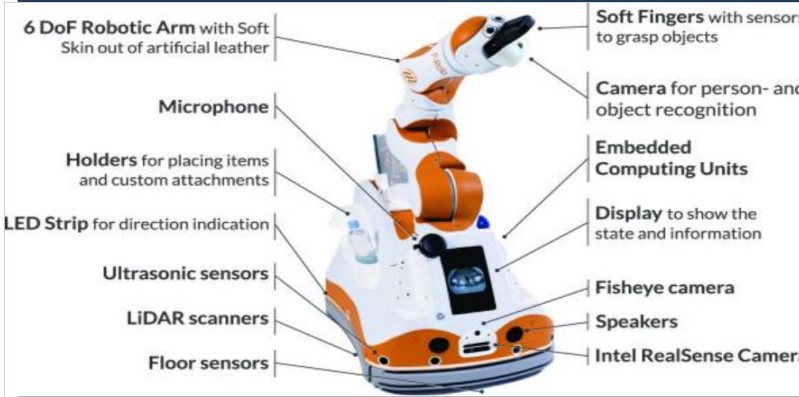


Fig. 4. Overview of Lio: sensor and hardware setup.

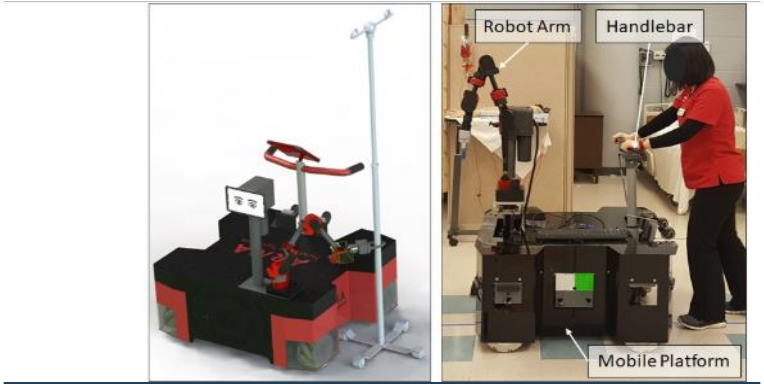


Fig. 5. Adaptive robotic nursing assistant including an omnidirectional mobile platform

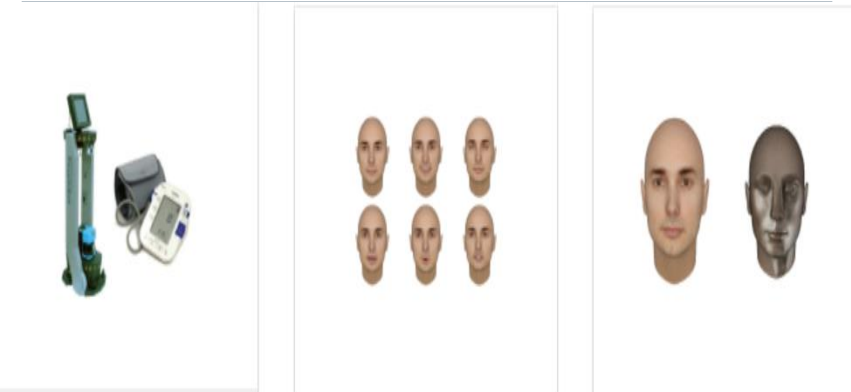


Fig. 6. Peoplebot robot and blood pressure monitor and human-like face created by Facegen



Fig. 7. Main stages of the home-care nursing controlled mobile robot

Research Gap



- **Limited Continuous Monitoring:**
Most robotic nurses lack continuous real-time vital monitoring.
- **Prototype-Level Systems:**
Many solutions remain at prototype stage without hospital validation.
- **No Early Prediction or Alerts:**
Few systems provide real-time abnormality detection or alerts.
- **Weak Human–Robot Interaction:**
Limited interaction reduces patient trust and acceptance.
- **Poor Workflow Integration:**
Not well integrated with nurse workflows or hospital systems.
- **High Cost and Complexity:**
Expensive and complex systems limit wider adoption.
- **Limited Task Agility:**
Struggle with dynamic hands-on caregiving tasks.
- **Limited Data Analytics:**
Lack advanced analytics and personalized health insights.

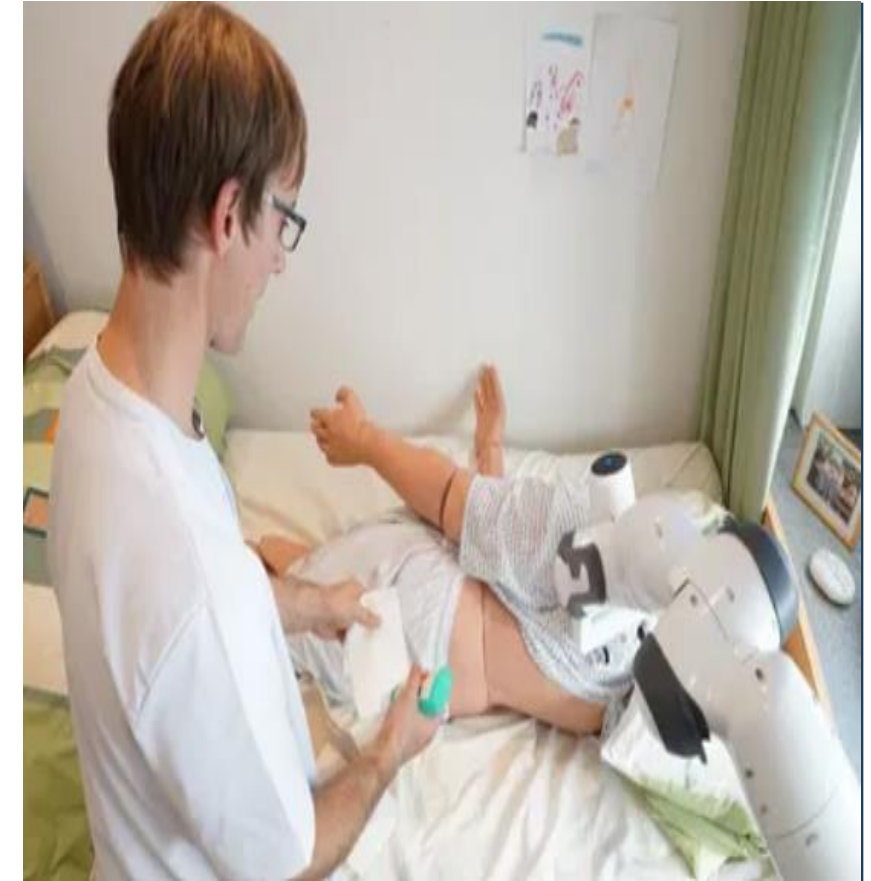


Fig.8. Source: www.google scholar.com

Research Objectives



1. Design & Development

- Robotic structure with sensors for vital monitoring (ECG, SpO₂, skin temp)
- Embedded electronics & mechanical design for patient assistance Patient–clinician interaction interface

2. Mathematical Modeling

- Multi-dimensional movement & control modeling
- Compensation for patient motion

3. Simulation & Analysis

- Task simulation before deployment
- Analysis under varied patient & environment conditions
- Evaluation of human–robot interaction and acceptance



Fig. 9.

Source: www.google scholar.com



- **SDG 3 – Good Health and Well-being**
 - Nuria improves patient monitoring, early detection of abnormal conditions, and support for caregivers, contributing directly to better healthcare outcomes and patient safety.
- **SDG 9 – Industry, Innovation, and Infrastructure**
 - Developing an intelligent robotic nurse promotes innovation in healthcare technology and builds infrastructure for smart hospital automation.

Plan of action



Milestones/Deliverables	Semester 4/6					
	Dec 25	Jan 26	Feb 26	Mar 26	April 26	May 26
Literature Survey, Objective, Resource collection						
System Architecture & Conceptual Design						
Mathematical Modeling & Control Design						
Simulation and Paper 1 st Draft						
Conference/Publication Communication						
Documentation						

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